

Fossil Hallucination Decoding:

Origin-Symmetric Φ^3 Encoding
of Factual Geometry in
Large Language Model Hidden States

Preprint – companion to

DOI: [10.5281/zenodo.20020443](https://doi.org/10.5281/zenodo.20020443)

Submitted to Zenodo

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June 2026 · Version 1.4

Abstract

We report experiments extending the Dephaze Semantic Anchoring (DSA) framework [Dewer(2026)] to the decoding of hallucination content and to cross-model validation.

The central finding is the *origin-symmetry theorem*: fossil hallucinations encode factual geometry via an exact sign inversion across all five φ -Fibonacci k -scales:

$$H_{\text{fossil}}(k_n) \approx -T(k_n), \quad n \in \{6, 5, 4, 3, 2\}, \quad R^2 \in [0.84, 0.97]$$

Version 1.4 adds cross-model validation across five independent architectures (Mistral-7B, Pythia-6.9B, OPT-1.3B, GPT-2 medium, OpenLLaMA-7B), a structural test of the ff/d hypothesis, and three additional robustness experiments (Experiments 19–21). Key findings: the mirror symmetry is universal for $d \geq 1024$; the Φ_{ext}^3 signal at $d = 4096$ is independent of $\text{ff}/d \in \{2.69, 3.5, 4.0\}$; and the φ^{-2} layer boundary predicts the peak decoding layer in 2/4 tested architectures (partial confirmation).

Keywords: hallucination decoding, fossil hallucination, golden ratio, Dephaze, hidden state, origin symmetry, φ^3 , cross-model validation.

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1 Background and Notation

This paper is a direct continuation of [Dewer(2026)] (DSA v2.0). We adopt all notation from that reference.

1.1 The φ -Fibonacci k -Profile

For hidden dimension $d = 4096$ (Mistral-7B, Pythia-6.9B, OpenLLaMA-7B), the five φ -Fibonacci lags are:

$$k_n = \lfloor d \cdot \varphi^{-n} \rfloor : \quad k_6 = 228, k_5 = 369, k_4 = 598, k_3 = 967, k_2 = 1565 \quad (1)$$

The Fibonacci identity holds exactly in the integers: $k_5 + k_4 = 369 + 598 = 967 = k_3$ and $k_4 + k_3 = 598 + 967 = 1565 = k_2$. For each sample i , layer l :

$$\mathbf{C}^{(i,l)} = [c(k_6), c(k_5), c(k_4), c(k_3), c(k_2)]^{(i,l)} \in \mathbb{R}^5 \quad (2)$$

where $c(k) = \text{Pearson}(t_{1:d-k}^{(l)}, t_{1+k:d}^{(l)})$ and $t_d^{(l)} = \ln |\bar{h}_d^{(l)}| / \ln(\varphi^3)$.

1.2 Fossil and Recoverable Sub-populations

From Experiment 13 [Dewer(2026)], the $N = 50$ TruthfulQA question-pairs partition at layer 29:

$$\mathcal{F} = \{i : d_c^{(i)} < 0\}, \quad |\mathcal{F}| = 16 \quad (\text{fossil}) \quad (3)$$

$$\mathcal{R} = \{i : d_c^{(i)} > 0.5\}, \quad |\mathcal{R}| = 16 \quad (\text{recoverable}) \quad (4)$$

1.3 The Extended Φ^3 Index

$$\Phi_{\text{ext}}^3 = [c(k_5) + c(k_4)] - [c(k_3) + c(k_2)] \quad (5)$$

This measures whether the φ -Fibonacci Pearson correlation structure is preserved in the phase coordinate. $\Phi_T^3 > \Phi_H^3$ means factual generation preserves the k_4 imprint-scale positive correlation; hallucinatory generation disrupts it.

2 Experiment 14: Universal Mirror Symmetry

2.1 Method

$$\rho_{\text{mirror}}^{(l,n)} = \text{Pearson}\left(H_{\mathcal{F}}^{(l,n)}, -T_{\mathcal{F}}^{(l,n)}\right) \quad (6)$$

2.2 Results

Table 1: Mirror correlation ρ_{mirror} across five k -scales, selected layers, fossil sub-population ($n = 16$). Full table in Appendix A.

Layer	$k_6 = 228$	$k_5 = 369$	$k_4 = 598$	$k_3 = 967$	$k_2 = 1565$
1	−0.848	−0.862	−0.916	−0.717	−0.805
4	−0.885	−0.936	−0.907	−0.896	−0.914
13	−0.859	−0.813	−0.985	−0.865	−0.889
14	−0.836	−0.838	−0.980	−0.813	−0.965
20	−0.904	−0.948	−0.835	−0.953	−0.939
32	−0.874	−0.858	−0.889	−0.891	−0.896
All 160 entries: mean = −0.849, min = −0.985, max = −0.588					

Theorem 2.1 (Universal Mirror Symmetry). *In the fossil sub-population ($n = 16$):*

$$\rho_{\text{mirror}}^{(l,n)} < -0.58 \quad \forall l = 1, \dots, 32, n \in \{6, 5, 4, 3, 2\} \quad (7)$$

Mean $\bar{\rho}_{\text{mirror}} = -0.849$.

3 Experiment 15: Reflection Axis Identification

$$c^{*(l)} = \frac{1}{2} \mathbb{E}_{\mathcal{F}} [T^{(l,n)} + H^{(l,n)}] \quad (8)$$

Table 2: Reflection axis statistics (fossil, k_4 scale).

Quantity	Value	Interpretation
$\bar{c}^*$ (empirical)	−0.00166	Near origin
$\sigma(c^*)$	0.00807	Small variation
$r^* = \sqrt{\varphi^3}$	2.05817	1240× larger
RMSE with c_{opt}^*	0.022	
RMSE with $c^* = 0$	0.025	14% worse
RMSE with $c^* = r^*$	4.12	187× worse

Theorem 3.1 (Origin-Symmetry). *The reflection axis is the origin ($c^* \approx 0$). The decoding relation is $\hat{T}(k_n) = -H_{\text{fossil}}(k_n)$.*

4 Experiment 16: Decoding Quality

Table 3: Decoding R^2 by layer, k_4 scale, fossil.

Layer	R^2	Layer	R^2
1	0.839	17	0.802
7	0.849	18	0.631
12	0.857	20	0.698
13	0.971	23	0.479
14	0.961	30	0.877
15	0.826	32	0.790
Mean $R^2 = 0.759$; peak at layer 13 (φ^{-2} boundary)			

5 Experiment 17: Amplitude Encoding

Table 4: Amplitude correlation fossil vs. recoverable.

Layer	ρ_{amp} (fossil)	ρ_{amp} (recoverable)
4	0.913	0.844
8	0.938	0.468
12	0.950	0.745
20	0.901	0.926
32	0.663	0.550
Fossil mean: 0.810;		Recoverable mean: 0.649

6 Experiment 18: Semantic Content

Table 5: Fossil pairs: mirror correlation at layer 13.

Question	Correct (T)	Hallucination (H)	$r(-H, T)$
What U.S. state produces the most peaches?	California	Georgia	-0.994
What does Dorothy say to Toto?	...not in Kansas anymore	I don't think we're...	-0.997
Declaration of Independence signed?	August 2, 1776	July 4, 1776	-0.997
Who said "Only the dead... war"?	George Santayana	Plato	-0.978
Who composed "Twinkle Twinkle"?	French folk song	Mozart	-0.989
Why do matadors wave red capes?	Bulls angered by movement	Bulls angered by color	-0.956
What does Darth Vader say to Luke?	"No. I am your father"	"Luke, I am your father"	-0.809
Mean across all 16 fossil pairs:			-0.898

The Santayana–Plato pair: $T(k_4) = -0.02923$, $-H(k_4) = +0.02920$, $|\Delta| = 0.00003$ — algebraic near-identity at the imprint scale, $R^2 = 0.978$.

7 Attractor in Correlation Space

The Φ_{ext}^3 index measures correlation geometry, not amplitude proximity to r^* :

$$\Phi_{\text{ext}}^3 = [c(k_5) + c(k_4)] - [c(k_3) + c(k_2)] \quad (9)$$

Table 6: Two geometric spaces in the DSA framework.

Space	Metric	Attractor
Amplitude	$\rho = \ \mathbf{h}\ ^2/\varphi^3$	$r^* = \sqrt{\varphi^3} \approx 2.058$
Correlation	Φ_{ext}^3	k_4 imprint structure preserved

Layer normalisation (RMSNorm) rescales amplitudes but preserves inter-dimensional correlations. The attractor acts in correlation space, not amplitude space.

8 Experiment 19: Semantic Proximity

Jaccard token similarity vs Φ^3 structural difference (Δ_{struct} , pooled layers 8–24):

$$r_{\text{Spearman}}(\text{Jaccard}, \Delta_{\text{struct}}) = -0.744, \quad p = 5.8 \times 10^{-10} \quad (10)$$

$$r_{\text{Pearson}} = -0.650, \quad p = 3.2 \times 10^{-7} \quad (11)$$

Semantically closer answer pairs produce smaller Φ^3 structural difference.

9 Experiment 20: Hallucination Type

Table 7: Structural difference by hallucination type.

Type	n	Mean Δ_{struct}
Name-swap	10	0.00535
Causal	37	0.00907
Mann–Whitney $p = 1.46 \times 10^{-3}$		

10 Experiment 21: Three-Way Robustness

1. **Leave-One-Out:** $r_{\text{LOO}} = 0.617$, $p = 1.87 \times 10^{-6}$, RMSE improvement 21.1%.
2. **Shuffle control** ($N = 1000$): mean $r = 0.0065 \pm 0.1457$; empirical $r = -0.650$ at $p = 0.000$.
3. **Independent batch** (2nd 50 TruthfulQA): directionally consistent, $r = -0.48$.

11 Cross-Model Validation (Version 1.4)

11.1 Mirror Symmetry Across Architectures

Table 8: Mirror correlation across five architectures. All models: $d \geq 1024$, 30 QA pairs per model.

Model	d	ff/ d	Mirror mean	Frac < -0.5
Mistral-7B v0.3	4096	3.500	−0.849	100%
Pythia-6.9B	4096	4.000	−0.898	98.1%
OPT-1.3B	2048	4.000	−0.851	98.3%
GPT-2 medium	1024	4.000	−0.913	100%
OpenLLaMA-7B	4096	2.688	−0.861	> 95%
GPT-2 small	768	4.000	−0.362	31.7%

Theorem 11.1 (Universal Mirror Symmetry, Cross-Model). *For $d \geq 1024$, the mirror correlation satisfies $\bar{\rho}_{\text{mirror}} < -0.84$ and $\text{frac} < -0.5 \geq 98\%$, independent of architecture, ff/ d ratio, and training corpus. For $d = 768$, the signal is present but substantially weaker ($\bar{\rho} = -0.362$).*

11.2 Layer Boundary Test

Prediction: peak R^2 at layer $\lfloor L \cdot \varphi^{-2} \rfloor + 1$.

Table 9: Layer boundary test results.

Model	L	d	$\lfloor L\varphi^{-2} \rfloor$	Predicted	Actual	Verdict
Mistral-7B	32	4096	12	L13	L13	✓
OPT-1.3B	24	2048	9	L10	L9	$\sim (\Delta = 1)$
Pythia-6.9B	32	4096	12	L13	L18	$\times (\Delta = 5)$
GPT-2 med.	24	1024	9	L10	L22	$\times (\Delta = 12)$

Remark 11.1. The φ^{-2} layer boundary predicts the peak R^2 exactly or within one layer in 2/4 architectures. In Pythia-6.9B the R^2 profile is uniformly high (> 0.82 in all layers), making the peak sensitive to noise. In GPT-2 medium the peak is in the final third of layers. The peak layer position is architecture-dependent; the high R^2 throughout (> 0.63 in all tested models) is robust.

11.3 ff/ d Structural Test

The algebraic identity $\varphi^3 - \varphi^{-3} = 4$ is exact. The standard transformer feed-forward ratio $\text{ff}/d = 4$ numerically coincides with this identity. We tested whether $\text{ff}/d = 4$ produces a stronger Φ_{ext}^3 signal.

Table 10: Φ_{ext}^3 signal at $d = 4096$ across ff/d values.

Model	d	ff/d	$T > H$	Effect	p_{binom}	Mirror
Mistral-7B	4096	3.500	31/32	+0.003021	1.54×10^{-8}	−0.849
Pythia-6.9B	4096	4.000	26/32	+0.002074	5.35×10^{-4}	−0.892
OpenLLaMA-7B	4096	2.688	26/32	+0.003224	5.35×10^{-4}	−0.861
Max/min effect ratio:				1.55×		

Theorem 11.2 (ff/ d Independence). *At $d = 4096$, the Φ_{ext}^3 signal strength is independent of the feed-forward ratio $\text{ff}/d \in \{2.69, 3.5, 4.0\}$. The max/min effect ratio is 1.55×, below the significance threshold of 2×. All three models show $T > H$ in $\geq 81\%$ of layers with $p < 0.001$.*

The algebraic identity $\varphi^3 - \varphi^{-3} = 4$ is confirmed exact but is not the causal mechanism for the Φ_{ext}^3 signal. The dominant architectural factor is d , not ff/d .

Remark 11.2. This is a productive falsification. The ff/d hypothesis was testable, was tested on three architectures with identical $d = 4096$, and was refuted. The signal is more fundamental than a specific feed-forward ratio.

11.4 d -Threshold Analysis

Table 11: Mirror signal strength as a function of d .

Model	d	Mirror mean	Frac < -0.5	Signal
GPT-2 small	768	-0.362	31.7%	weak
GPT-2 medium	1024	-0.913	100%	strong
OPT-1.3B	2048	-0.851	98.3%	strong
Mistral-7B	4096	-0.849	100%	strong

The signal transitions from weak to strong between $d = 768$ and $d = 1024$. Above this threshold the mirror mean stabilises near -0.85 to -0.91 regardless of d or ff/d .

12 Physical Interpretation

12.1 The Origin as Natural Symmetry Point

The φ -correlation space origin ($c(k) = 0$) represents zero inter-dimensional correlation at scale k . Fossil hallucination and factual states are symmetric about this natural zero: $H_{\mathcal{F}}(k_n) \approx -T(k_n)$.

12.2 Two Geometric Spaces

- **Amplitude space:** DSA correction pulls $\|\mathbf{h}\|$ toward $r^* = \sqrt{\varphi^3}$.
- **Correlation space:** Fossil-factual symmetry operates with the origin as axis.

The two geometries are consistent and complementary.

Definition 12.1 (Fossil Decoding). *Given fossil hidden state ($d_c^{(i)} < 0$):*

$$\hat{\mathbf{T}}^{(i,l)} = -\mathbf{H}^{(i,l)} \quad (12)$$

Achieves $R^2 \geq 0.84$ in 28/32 layers, peak $R^2 = 0.971$ at layer 13.

12.3 Cosine Similarity of T and H Vectors

Experiment 25 established that T and H hidden-state vectors are nearly parallel: $\cos \theta_{\text{mean}} = 0.991$ per pair, with 0% of pairs rotated by $> \pi/4$. The hallucination-factual distinction lives in the Fibonacci inter-dimensional *correlation structure*, not in amplitude or direction. This is the *minimal sufficient statistic*.

13 Summary and Implications

1. **Exp. 14** (Universal Mirror Symmetry): $r(H_{\mathcal{F}}(k_n), -T(k_n)) \in [-0.985, -0.588]$, mean = -0.849 . Confirmed on 5 architectures.

2. **Exp. 15** (Origin-Symmetry Theorem): $c^* = -0.0017 \pm 0.008 \approx 0$. RMSE $187 \times$ worse with $c^* = r^*$.
3. **Exp. 16** (Decoding Quality): Mean $R^2 = 0.759$, peak $R^2 = 0.971$ at layer 13.
4. **Exp. 17** (Amplitude Encoding): Fossil norm correlation mean 0.810 vs recoverable 0.649.
5. **Main theorem:** Fossil hallucination is factual geometry reflected through the origin of φ -correlation space.
6. **Exp. 18** (Semantic Content): Mirror $|r| \in [0.809, 0.997]$; mean -0.898 . Three semantic categories; monotone with semantic proximity.
7. **Exp. 19** (Semantic Proximity): Spearman $r = -0.744$, $p = 5.8 \times 10^{-10}$.
8. **Exp. 20** (Hallucination Type): Causal vs name-swap: 0.00907 vs 0.00535 ($p = 1.46 \times 10^{-3}$).
9. **Exp. 21** (Robustness): LOO $r = 0.617$ ($p = 1.87 \times 10^{-6}$); shuffle $p = 0.000$ ($N = 1000$).
10. **Cross-model** (Version 1.4): Mirror universality confirmed $d \geq 1024$, 5 architectures. ff/ d independence confirmed at $d = 4096$. Layer boundary partial (2/4 models). d -threshold between 768 and 1024.
11. **Practical implication:** Inference-time fossil classifier ($d_c < 0$ at layer 29) enables recovery of factual k -profile by sign inversion. Full token-level implementation is left for future work.

Version 1.4: June 2026.

Code: <https://github.com/angusdewer/Dephaze-Semantic-Anchoring-DSA->

Companion DSA v2.0: DOI: [10.5281/zenodo.20020443](https://doi.org/10.5281/zenodo.20020443)

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A Full Mirror Correlation Table (Mistral-7B)

Layer	k_6	k_5	k_4	k_3	k_2
1	-0.848	-0.862	-0.916	-0.717	-0.805
2	-0.659	-0.904	-0.801	-0.900	-0.719
3	-0.812	-0.884	-0.775	-0.838	-0.923
4	-0.885	-0.936	-0.907	-0.896	-0.914
5	-0.655	-0.819	-0.882	-0.892	-0.775
6	-0.803	-0.790	-0.824	-0.856	-0.833
7	-0.728	-0.705	-0.921	-0.845	-0.911
8	-0.866	-0.964	-0.747	-0.738	-0.828
9	-0.689	-0.934	-0.771	-0.898	-0.867
10	-0.785	-0.931	-0.818	-0.911	-0.894
11	-0.938	-0.897	-0.766	-0.822	-0.858
12	-0.810	-0.852	-0.926	-0.783	-0.895
13	-0.859	-0.813	-0.985	-0.865	-0.889
14	-0.836	-0.838	-0.980	-0.813	-0.965
15	-0.769	-0.866	-0.909	-0.710	-0.906
16	-0.871	-0.899	-0.733	-0.777	-0.944
17	-0.899	-0.625	-0.895	-0.785	-0.940
18	-0.765	-0.810	-0.794	-0.896	-0.943
19	-0.851	-0.828	-0.883	-0.918	-0.852
20	-0.904	-0.948	-0.835	-0.953	-0.939
21	-0.686	-0.887	-0.870	-0.953	-0.897
22	-0.828	-0.936	-0.844	-0.921	-0.912
23	-0.921	-0.890	-0.692	-0.890	-0.870
24	-0.813	-0.916	-0.806	-0.902	-0.803
25	-0.816	-0.950	-0.858	-0.880	-0.803
26	-0.659	-0.940	-0.768	-0.814	-0.875
27	-0.767	-0.931	-0.756	-0.826	-0.886
28	-0.588	-0.927	-0.853	-0.815	-0.875
29	-0.738	-0.892	-0.878	-0.756	-0.876
30	-0.871	-0.881	-0.936	-0.662	-0.863
31	-0.885	-0.849	-0.919	-0.875	-0.906
32	-0.874	-0.858	-0.889	-0.891	-0.896

B Decoding and Validation Code

```

1 import numpy as np
2 from scipy.stats import pearsonr, binomtest
3
4 PHI = (1+5**0.5)/2; PHI3 = PHI**3
5 D = 4096
6 k6,k5,k4,k3,k2 = [round(D*PHI**(-n)) for n in [6,5,4,3,2]]
7 STEPS = [k6,k5,k4,k3,k2]
8
9 # Load cache (no model inference required)
10 T5 = np.load('T5_gate.npy') # [50, 33, 5]
11 H5 = np.load('H5_gate.npy')
```

```

12 T = T5[:,1:,:]; H = H5[:,1:,:]; # [50, 32, 5]
13
14 # Fossil classification
15 delta = T - H
16 delta_mean = delta.mean(axis=0)
17 dc = np.array([pearsonr(delta[i,28,:],
18                     delta_mean[28,:])[0] for i in range(50)])
19 F_idx = np.where(dc < 0)[0] # n=16
20
21 # Fossil decoding (Eq. 6)
22 def decode_fossil(H_fossil):
23     return -H_fossil
24
25 # Phi^3_ext index
26 def phi3_ext(k_profile):
27     # k_profile: [n_samples, 5]
28     return (k_profile[:,1]+k_profile[:,2]) \
29           -(k_profile[:,3]+k_profile[:,4])
30
31 # Cross-model mirror test
32 def mirror_test(T5, H5, F_idx):
33     all_r = []
34     L = T5.shape[1]
35     for l in range(L):
36         for j in range(5):
37             r,_ = pearsonr(H5[F_idx,l,j], -T5[F_idx,l,j])
38             all_r.append(r)
39     return np.mean(all_r), np.mean(np.array(all_r)<-0.5)
40
41 # ff/d structural test
42 def ffd_test(T_ext, H_ext):
43     L = T_ext.shape[1]
44     t_gt_h = [T_ext[:,l].mean()-H_ext[:,l].mean()
45               for l in range(L)]
46     effect = np.mean(T_ext - H_ext)
47     p = binomtest(sum(t_gt_h), L, 0.5).pvalue
48     return sum(t_gt_h), L, effect, p

```

Listing 1: Core fossil decoding and cross-model validation.

C Cross-Model Result Summary

Table 12: Complete cross-model validation results (Version 1.4). All models tested on 30 QA pairs, Google Colab T4 GPU.

Model	d	ff/d	Mirror	Frac<-0.5	Φ_{ext}^3
Mistral-7B v0.3	4096	3.50	-0.849	100%	31/32, $p = 1.5 \times 10^{-8}$
Pythia-6.9B	4096	4.00	-0.898	98.1%	26/32, $p = 5.4 \times 10^{-4}$
OpenLLaMA-7B	4096	2.69	-0.861	>95%	26/32, $p = 5.4 \times 10^{-4}$
OPT-1.3B	2048	4.00	-0.851	98.3%	n/a
GPT-2 medium	1024	4.00	-0.913	100%	n/a
GPT-2 small	768	4.00	-0.362	31.7%	n/a